Speaker Notes

What Near-Perfect Accuracy Does and Does Not Show on the UCI TCGA Pan-Cancer RNA-Seq Dataset: Leakage, Equivalence, Conformal Reliability and Data Efficiency

NIELIT Bio-AI Conclave 2026 · Track 2 · Er. Rishabh Aryan, IIIT Bhagalpur

Timing: about 12 minutes for 16 slides (roughly 45 s per content slide).

Slide 1 / 16 — Title

Good morning. I am Rishabh Aryan from IIIT Bhagalpur. This talk re-examines a popular benchmark for tumour-type classification from RNA-Seq data. Instead of proposing yet another model, I ask what the near-perfect accuracies reported on it can and cannot tell us: how much leakage matters here, whether any model is really better, and what the dataset can still say about reliability and data needs.

Sources: Title: paper title (PanCancer_paper_camera_ready.docx). Presenter, affiliation, e-mail, ORCID: as supplied by the presenter. Conclave, Track 2, dates and venue: NIELIT Bio-AI Conclave 2026 call for papers.

Slide 2 / 16 — Outline

The talk has six parts: the motivation and dataset with three research questions; the leakage-free methodology; the leakage audit, performance, all pairwise comparisons, and the errors with a reliability analysis using conformal prediction; then how much data the task needs, how leakage grows as data shrink, and selection stability; the limitations; and the takeaways.

Sources: Section structure follows the paper.

Slide 3 / 16 — 1. Introduction: Motivation

Knowing where a tumour comes from guides treatment, and it is hardest for cancers of unknown primary site. Gene expression carries a strong cell-of-origin signal, as the TCGA PanCancer Atlas showed. The UCI subset of TCGA is a popular benchmark, and accuracies near one hundred per cent are routinely reported, often with genes selected on the full dataset and without any statistical test. That is a known source of bias in small studies, but nobody has measured how large it is on this dataset. The figure shows that even two principal components separate the five tumour types quite well; this is a visual only, not a classifier.

Sources: [1] Fiorini 2016 UCI dataset; [2] Hoadley et al. 2018 Cell; [3] Ambroise & McLachlan 2002 PNAS; [5] Pavlidis & Pentheroudakis 2012 Lancet; [6] Kapoor & Narayanan 2023 Patterns; [7] Nitta et al. 2021 IEEE BigData. Figure: deck/figs/s_pca.png (same PCA as paper Fig. 1b).

Slide 4 / 16 — 1. Introduction: The Dataset

The data are the UCI 'gene expression cancer RNA-Seq' set: 801 TCGA tumours of five types, with 20,531 expression values each. The classes are unbalanced; colon adenocarcinoma is the smallest with 78 samples. The values are already on a log scale, 267 genes are constant, and nothing is missing. Importantly, the genes are named gene_0 to gene_20530; the repository says these are dummy names, so no gene symbols are available, and I do not interpret individual genes biologically. There are no clinical or batch variables.

Sources: [1] UCI dataset page https://archive.ics.uci.edu/dataset/401 (accessed 16 Oct 2026); counts and value range computed by run_all.py / data check.

Slide 5 / 16 — 1. Introduction: Research Questions

I ask three questions. First, how much does selecting genes on the full data actually change accuracy on this dataset? Second, do any of seven standard classifiers really differ, and which ones can be shown to be equivalent? Third, applied here for the first time as far as I know: are the predictions reliable, meaning calibrated and able to abstain when unsure, and how much data does the task really need? Common practice selects genes on all samples and reports a single split; here selection happens inside each training fold, the leaky variant is run as a deliberate comparison, and all twenty-one pairs of classifiers are tested with Holm adjustment, equivalence tests and a Bayesian test.

Sources: Research questions: paper Section 1. Protocol: paper Section 3. [3] Ambroise & McLachlan 2002; [4] Nadeau & Bengio 2003.

Slide 6 / 16 — 2. Methodology

Every model is wrapped in a pipeline that is refitted on each training fold: constant genes are dropped, the one hundred genes with the highest ANOVA F-statistic are kept, and the data are standardised with training statistics only. I compare softmax regression, ridge, a linear SVM, Gaussian naive Bayes, a random forest, LightGBM and a small MLP, plus a majority-class baseline, all with fixed default settings. Evaluation is ten-times repeated stratified five-fold cross-validation, fifty folds, identical for every model. Confidence intervals and paired tests use the Nadeau–Bengio correction for overlapping training sets. All twenty-one pairs are tested with Holm adjustment, plus two one-sided equivalence tests with a one-point margin and a Bayesian correlated t-test.

Sources: Paper Section 3.2–3.4; code src/experiments.py. [4] Nadeau & Bengio 2003 Machine Learning.

Slide 7 / 16 — 3. Results: Leakage Audit

Here is the central finding. I ran the same pipeline twice: once with gene selection and scaling fitted on all samples before cross-validation, the leaky way, and once inside each fold. With real labels and all 801 samples, the two are practically identical: 99.50 versus 99.48 per cent with one hundred genes. Only with ten genes is the leaky estimate visibly higher, and even that is not significant. With shuffled labels the leaky pipeline does score above the in-fold one, and in small samples of sixty tumours it reaches about 68 per cent on pure noise, which is the classic Ambroise and McLachlan effect. So on this dataset leakage hides behind the ceiling: published accuracies are probably not much inflated, but an accuracy near one hundred per cent cannot show that a pipeline is clean. A permutation test can.

Sources: Paper Section 4.2, Table 3, Fig. 1; results/leak_real_summary.csv, leak_perm_fullN.csv, leak_small_n.csv. [3] Ambroise & McLachlan 2002.

Slide 8 / 16 — 3. Results: Leakage-Free Performance

All seven classifiers reach between 98.9 and 99.8 per cent macro-F1, against 10.9 per cent for the majority baseline. Linear SVM is highest at 99.81 per cent, about 1.5 errors out of 801 per repeat; softmax regression reaches 99.48 per cent with about 4.2 errors. Macro ROC-AUC is at least 0.9944 for every model, so it does not discriminate between them. One error costs about 0.6 point in a fold of 160, so the differences are a few samples.

Sources: Paper Table 2; Supplementary S2; results/summary.csv.

Slide 9 / 16 — 3. Results: All Pairwise Comparisons

All models see the same fifty folds, so I compared every pair, twenty-one in total, with the corrected t-test and Holm adjustment. No pair differs significantly; the smallest adjusted p-value is 0.66. But no significant difference is not the same as no difference, so I also ran equivalence tests: 10 of the fifteen pairs that do not involve naive Bayes are equivalent within plus or minus one percentage point. Naive Bayes is probably slightly worse, but not significantly. The design can only detect differences of about 0.5 to 1.4 points, which is several samples per repeat. So the supported claim is no winner, with equivalence shown for most pairs; the remaining pairs are inconclusive, not different.

Sources: Paper Section 4.3; Supplementary Section S5; results/pairwise_all.csv. [4] Nadeau & Bengio 2003.

Slide 10 / 16 — 3. Results: Errors and Reliability

Softmax made 42 errors in 8,010 predictions, all from 5 samples; the three hard LUAD tumours have atypical expression profiles. Does the model know when it may be wrong? Its expected calibration error is 0.47 per cent, compared with about 0.14 per cent for a perfectly calibrated model with the same confidences, so the miscalibration is small and probably real, and it sits in the few low-confidence predictions. Split-conformal sets, calibrated inside each training fold, are valid on average, but at 90 and 95 per cent they reach coverage by returning empty sets, and fold coverage varies widely because only about 160 tumours are used for calibration. At 99 per cent, 99.4 per cent of samples get one label, and abstaining on the rest lowers the error among accepted predictions from 0.51% [0.42, 0.60] to 0.18% [0.10, 0.25]. A plain probability threshold does about as well, at 0.26 per cent, so what conformal adds is a threshold chosen with a coverage guarantee, not better error selection. The hard LUAD samples are flagged in 18 of their 30 appearances, about half. Class-conditional, Mondrian sets would need at least 99 calibration tumours in every class to work at 99 per cent; no class has that many here.

Sources: Paper Sections 4.4 and 4.6, Table 4, Fig. 2; Supplementary S7, S11; results/nov_conformal_.csv, rob_.csv. Conformal prediction: Vovk et al. 2022; Angelopoulos & Bates 2023. Calibration reference: Bröcker & Smith 2007.

Slide 11 / 16 — 4. Results: How Much Data?

How much data does this task need? Very little. With three training tumours per class, softmax reaches 98.1 per cent macro-F1 on a held-out fold, and with ten per class 99.4 per cent, plus or minus 0.2; naive Bayes needs far more. The right panel links this to the leakage audit: on shuffled labels, leaky selection inflates accuracy by 65 ± 5 points at thirty samples but 14 ± 2 at three hundred and ninety, while with real labels the gap stays near zero, and with ten genes it even turns slightly negative. This is the behaviour Vabalas and colleagues described in simulations, and learning curves for expression data go back to Mukherjee and colleagues; what is new here is only the numbers for this dataset. When ten tumours per class reach the ceiling, selection bias has no room to show.

Sources: Paper Section 4.7, Fig. 3; Supplementary S3, S12; results/rob_learning_curve_ci.csv, rob_leak_vs_n_ci.csv. [9] Vabalas et al. 2019; Mukherjee et al. 2003.

Slide 12 / 16 — 4. Results: Selection Stability

Across the fifty training folds, 133 genes entered the top one hundred at least once, and 76 were chosen every time. The average overlap between two folds' selections is a Jaccard index of 0.87, and the chance-corrected Nogueira stability is 0.93. A recent study by Sirohi on a larger TCGA extract found a 75-gene random-forest panel much less stable, Jaccard 0.32; the methods differ, so the lesson is that stability depends on the pipeline and should be measured. The genes here have dummy names, so I make no biological claims about them.

Sources: Paper Section 4.5; Supplementary S8; results/selection_stability.json, results/stability_phi.json. [8] Sirohi 2026 NHSJS.

Slide 13 / 16 — 5. Discussion: Limitations

What remains valid is the size of the leakage effect, the equivalence of the models and the selection stability, on this dataset, within TCGA, for five well-separated tissues. The dataset is small and single-source with no clinical, batch or gene-symbol information, and even simple global statistics of each profile predict the cancer type about half the time, so classes differ in overall profile properties too. I did not test an external cohort or metastatic samples. The statistical tests are approximate because fold scores are bounded at one hundred per cent, and the audit covers gene selection and scaling, not oversampling or PCA.

Sources: Paper Section 7.

Slide 14 / 16 — 6. Conclusion: Takeaways and Future Work

To conclude: on this dataset at full size, selecting genes on all the data barely changes accuracy; the bias only shows with shuffled labels or small samples. All seven models reach 98.9 to 99.8 per cent macro-F1, none is significantly better, and 10 of the fifteen pairs without naive Bayes are demonstrably equivalent within one point. The reliability analysis shows small and probably real miscalibration, and 99 per cent conformal sets let the model abstain on under one per cent of samples while cutting the error among accepted predictions to 0.18 per cent, much as a probability threshold would; ten tumours per class are enough, which explains why leakage cannot show. So a near-perfect score here proves neither that a pipeline is clean nor that a model is better; use permutation and equivalence tests. Next steps are mapping gene indices to symbols, external validation, the full 33-type problem, and metastatic samples.

Sources: Paper Sections 5–7.

Slide 15 / 16 — References

These are the works cited on the slides; the paper has the full list.

Sources: Full references: paper reference list (all DOIs checked via Crossref, 16 Oct 2026).

Slide 16 / 16 — Thank You

Thank you for your attention. I am happy to take questions.

Sources: Contact details as supplied by the presenter.